

MEZRAIT LIUL MENBER

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Research interests: symmetric-key cryptography, cipher design & cryptanalysis, applied security

EDUCATION

Abu Dhabi University

B.Sc. in Cybersecurity Engineering — CGPA ██████████

Abu Dhabi, UAE

Expected June 2027

- Dean's List: **five consecutive** semesters (Fall 2023 – Fall 2025); full undergraduate scholarship awarded by the Government of the United Arab Emirates
- Relevant Coursework: Introduction to Cryptography; Network Security and Forensics; Ethical Hacking; Information Security; Digital Forensics; Data Structures and Algorithms; Discrete Mathematics; Statistics and Research Methods

PUBLICATIONS

AEX-128: A Key-Dependent Finite-State SPN–ARX Hybrid Block Cipher — *IEEE Access*, 2026

– M. Alawida, M. L. Menber — doi: [10.1109/ACCESS.2026.3730739](https://doi.org/10.1109/ACCESS.2026.3730739)

– A 128-bit cipher whose round structure is selected at runtime by a key-dependent finite-state machine, yielding **59,049** distinct execution paths — evaluated via MILP-based active-S-box analysis, SAC chi-square testing, and NIST SP 800-22 at 10^6 -bit scale

RESEARCH EXPERIENCE

Undergraduate Researcher, Block Cipher Design

Abu Dhabi, UAE

Abu Dhabi University — Supervisor: *Dr. Moatsum Alawida*

Feb 2026 – Present

- Co-designed AEX-128, a published 128-bit block cipher combining SPN and ARX rounds under a key-dependent finite-state machine — **3^{10} (59,049)** possible execution paths
- Designed the ChaCha-based key schedule: 512-bit state expansion via quarter-round mixing deriving 12 subkeys and per-round FSM transition parameters from key entropy
- Built the Java reference implementation used for the paper's benchmarks — **0.58 μ s/block** encryption — and designed the initial diffusion experiments, including the avalanche methodology retained in the published paper
- Co-developed the security evaluation with my supervisor — MILP-based active-S-box analysis across all 59,049 FSM execution paths, SAC testing with chi-square analysis (**94.5%** pass at 95% confidence), and NIST SP 800-22 testing at 10^6 -bit scale — and have since built an independent reproduction of the MILP analysis

TECHNICAL SKILLS

Cryptography & analysis: block cipher design, ChaCha-based key scheduling, avalanche/SAC evaluation, MILP active-S-box analysis (co-developed and independently reproduced), NIST SP 800-22

Languages: Java, Python; familiar: C, SQL, JavaScript/React

Tools: Git/GitHub, Linux, automated testing, LightRAG/knowledge-graph retrieval, digital-forensics tooling

PROFESSIONAL EXPERIENCE

Python Developer Intern

Abu Dhabi, UAE

Ansen Technology Co. Ltd. — AI-security company, eThreats threat-intelligence platform

Jun – Aug 2026

- Built the agent skill-rating lifecycle end-to-end, including the “Playbook Keeper” human-review interface and redesigned guidance delivery to specialist agents
- Improved agent tool-call reliability: traced and fixed the root cause of **94.1%** of failed tool calls through a live controlled experiment on production logs, and fixed a caching bug causing redundant calls in **294 of 385** runs — both shipped with regression tests
- Migrated the document-retrieval pipeline to knowledge-graph-enhanced retrieval (LightRAG) behind an adapter layer with zero-downtime shadow ingest — improved answer faithfulness from **86.7% to 100%** on a 15-query golden set and reported the ingest/latency trade-offs honestly

Course Assistant & Peer Mentor

Abu Dhabi, UAE

Abu Dhabi University

Jun 2024 – Present

- Delivered **50+ hours** of tutoring per semester across undergraduate mathematics, physics, chemistry, and engineering courses — certified for the 50-hour tutoring milestone in five consecutive semesters (**250+ hours** total)
- Mentored cohorts of **~10** first-year students each semester (**50+ overall**) through their academic and campus transition, recognized with the peer-mentorship certificate each term

PROJECTS

AEX-128 MILP Analysis — Independent Reproduction | *Python, PuLP/HiGHS*

code available on request

- Implemented the byte-truncated differential MILP model from scratch and exhaustively solved all **59,049** execution paths to proven optimality, with a solver-free dynamic-programming cross-verification
- Validated by independently reproducing the classical 5-round AES bound of **26** active S-boxes

Digital Forensics — NIST Data Leakage Case Study | *forensic tooling, full report*

[github](#)

- Conducted evidence recovery and investigation across the NIST data-leakage case using standard forensic tools; documented findings in a complete investigative report

HONORS & AWARDS

GISEC 2025: 5th place (of 10+ finalist teams), School of Cyber Defense CTF — Dubai Electronic Security Center; qualified 2nd among university teams in round one

Sharjah Voluntary Award: 2024 & 2025 consecutive recipient; 450+ hours of university community service

Certifications: Google Cybersecurity Certificate (2025); IBM Cybersecurity Professional Certificate (2025); Cisco CCNA: Introduction to Networks (2025); Cisco Introduction to Cybersecurity (2024)

Languages: Amharic (native); English (fluent)